

Quality-Aware Low-Resource Pashto Neural Machine Translation: Semantic Filtering, LoRA Adaptation, Multilingual Baselines, and Pivot-Based Hindi Translation

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github.com/Hrithikcrick/pashto-nmt-carnot-internship

Abstract—Low-resource neural machine translation is constrained not only by corpus size but also by duplication, weak alignment, domain mismatch, and unreliable evaluation. This paper presents a quality-aware Pashto–English translation pipeline using NLLB-200, semantic sentence-pair filtering, and LoRA parameter-efficient adaptation, together with direct and pivot-based Hindi extensions. Starting from 93,498 raw Pashto–English sentence pairs, rule-based cleaning retained 90,978 pairs. Standardized evaluation was conducted on the same 100 Pashto–English examples for the pretrained NLLB baseline, original LoRA, semantic-filtered LoRA, M2M100, and mBART-50. The pretrained NLLB model obtained BLEU 19.63, chrF++ 35.48, TER 75.68, and COMET 0.6818. Original LoRA obtained BLEU 19.77, chrF++ 36.31, TER 76.83, and COMET 0.6775. Semantic-filtered LoRA achieved the highest BLEU and chrF++ values among the NLLB variants at 20.38 and 36.61, respectively, while its COMET score of 0.6819 remained approximately tied with the pretrained baseline. The revised experimental package additionally introduces informative pipeline and metric visualizations, external multilingual baselines, qualitative translation examples, paired bootstrap analysis, automatic diagnostics, source-grouped leakage-controlled splits, dataset checksums, experiment manifests, and multi-seed configurations. The results support semantic filtering as a useful data-selection strategy for LoRA adaptation, but do not establish broad statistically significant superiority over the strong multilingual baseline.

Index Terms—Low-resource machine translation, Pashto, NLLB, LoRA, PEFT, Hugging Face, semantic filtering, BLEU, chrF, pivot translation, Hindi translation, IndicTrans2.

I. Introduction

Neural Machine Translation (NMT) has significantly improved translation quality for high-resource languages such as English, French, German, Spanish, Hindi, and Chinese. Large multilingual models can now translate between hundreds of languages with reasonable fluency. However, low-resource languages still face major limitations because they do not have sufficient high-quality parallel corpora. Pashto is one such language. Although Pashto has millions of speakers, the amount of publicly available, clean, and evaluation-ready parallel data is much smaller than what is available for high-resource language pairs.

The difficulty of Pashto translation is not only caused by limited dataset size. The quality of available data is equally important. Publicly available low-resource datasets often contain duplicated rows, very short sentence pairs, extremely long mismatched pairs, noisy web text, untranslated fragments, URL-heavy content, inconsistent script usage, and weak semantic alignment. If such data is used directly for training, the model may learn noisy mappings and produce incomplete, literal, or hallucinated translations.

This paper presents a structured Pashto NMT research pipeline that uses a pretrained multilingual translation model, performs dataset cleaning, evaluates baseline translation quality, prepares high-quality subsets, applies semantic similarity filtering, integrates manual error analysis, and then applies LoRA-based parameter-efficient fine-tuning. The baseline model used in this work is `facebook/nllb-200-distilled-600M`. The project focuses primarily on Pashto-to-English translation because English references are available for automatic evaluation. Direct Pashto-to-Hindi and pivot-based Pashto-to-English-to-Hindi translation are also included as extended translation directions.

The work is designed as a practical research pipeline rather than only a model training experiment. It includes dataset engineering, evaluation design, metric analysis, semantic filtering, error categorization, fine-tuning, Hindi comparison, and improvement planning. This is important because in low-resource NMT, model choice alone is not enough. Data quality, evaluation reliability, and error analysis strongly influence the final translation quality.

II. Motivation

Low-resource machine translation systems face several practical problems. A pretrained multilingual model may generate fluent output, but fluency does not always mean correctness. In Pashto translation, the model may omit important words, mistranslate named entities, change tense, produce overly literal English, or hallucinate content that is not present in the source sentence. These errors are especially harmful

when translation is used for research, education, information access, or downstream multilingual applications.

A second motivation is the need for Hindi translation. Direct Pashto-to-Hindi translation is difficult because high-quality Pashto-Hindi parallel data is limited. A pivot-based strategy can be more practical: Pashto is first translated into English, and then English is translated into Hindi. If the Pashto-to-English stage improves, the downstream Hindi translation can also improve.

A third motivation is computational practicality. Full fine-tuning of large multilingual models is expensive. LoRA provides a practical solution by training only a small number of adapter parameters while keeping most pretrained model weights frozen. This makes fine-tuning possible under limited compute and storage constraints.

III. Research Questions

This work is guided by the following research questions:

- How much noisy data can be removed from the raw Pashto-English dataset through rule-based cleaning?
- How strong is the pretrained NLLB model as a Pashto-to-English baseline?
- Can semantic similarity filtering improve the quality of the training subset?
- Can LoRA fine-tuning improve chrF over the baseline model?
- Does semantic-filtered LoRA perform better than original LoRA training?
- Why is manual error analysis necessary beyond BLEU and chrF?
- Is pivot-based Hindi translation more promising than direct Pashto-to-Hindi translation?
- What further steps are required to move toward deployment-level translation quality?

IV. Research Contributions

The revised study makes the following contributions:

- 1) A complete low-resource Pashto-English NMT pipeline based on NLLB-200 and LoRA.
- 2) Rule-based cleaning of a 93,498-pair corpus, retaining 90,978 parallel sentence pairs.
- 3) Semantic similarity filtering and filtered training subsets containing 8k, 4k, and 2k pairs.
- 4) Standardized comparison of pretrained NLLB, original LoRA, and semantic-filtered LoRA on the same test instances.
- 5) External zero-shot comparisons with M2M100 and mBART-50 [9], [10].
- 6) Evaluation using BLEU, chrF++, TER, and COMET [11].
- 7) Paired bootstrap confidence intervals and significance analysis.
- 8) Automatic diagnostics for number mismatch, output length, repetition, empty output, and exact reference matching.

- 9) Informative research-pipeline and model-comparison figures.
- 10) Qualitative baseline-versus-LoRA improvement and regression candidates.
- 11) Fixed source-grouped splits with zero source-level and exact-pair leakage.
- 12) Dataset hashes, environment records, training manifests, and reproducible experiment configurations.
- 13) Controlled configurations for seeds 13, 42, and 2026.
- 14) A successfully validated LoRA smoke-training and adapter-loading pipeline.
- 15) Direct and pivot Hindi analysis, with IndicTrans2 retained only for the applicable English-to-Hindi pivot stage.

V. Related Work

Multilingual NMT has become an important direction for low-resource languages because a single model can share representations across many languages. The No Language Left Behind (NLLB) model was developed to improve translation coverage for more than 200 languages, including many low-resource languages [1]. NLLB is suitable for this project because it supports Pashto, English, and Hindi with explicit language codes.

LoRA, or Low-Rank Adaptation, is a parameter-efficient fine-tuning method that freezes the base model and introduces trainable low-rank matrices into selected layers [2]. This approach reduces trainable parameter count and makes adaptation practical under limited compute. In this project, LoRA is applied to selected attention projection modules of the NLLB model.

BLEU is a widely used metric for machine translation evaluation [3]. It measures n-gram overlap between generated translations and references. However, BLEU can be harsh for morphologically rich and low-resource settings. chrF, based on character n-gram F-score, can better reflect partial lexical and morphological similarity [4]. Therefore, this project reports both BLEU and chrF.

VI. System Overview

The complete research pipeline consists of the following stages:

- 1) Collection of Pashto-English parallel data.
- 2) Conversion of raw data into structured CSV format.
- 3) Dataset cleaning and noise removal.
- 4) Train-validation-test split creation.
- 5) Baseline translation using NLLB.
- 6) Evaluation using BLEU, chrF, and inference time.
- 7) High-quality 10k subset construction.
- 8) Semantic similarity filtering.
- 9) Gold test candidate preparation.
- 10) LoRA-based fine-tuning.
- 11) Semantic-filtered LoRA fine-tuning.
- 12) Baseline versus fine-tuned checkpoint comparison.
- 13) Direct Hindi versus pivot Hindi comparison.
- 14) IndicTrans2 exploration for improved Hindi generation.

TABLE I
Dataset cleaning summary.

Stage	Rows
Raw sentence pairs	93,498
After duplicate removal	93,418
Final clean sentence pairs	90,978
Removed noisy rows	2,440

TABLE II
Train-validation-test split.

Split	Rows
Training	72,782
Validation	9,098
Test	9,098

15) Improved LoRA experiment planning.

VII. Dataset Preparation

A. Raw Dataset

The project uses Pashto-English parallel data converted into CSV format. The raw dataset contained 93,498 sentence pairs. Each row contains a Pashto source sentence and an English target sentence.

B. Cleaning Strategy

The cleaning pipeline removes duplicate, noisy, and weakly useful rows. The cleaning rules include:

- Removing empty rows.
- Removing duplicate sentence pairs.
- Removing URL-heavy and HTML-contaminated text.
- Removing very short sentence pairs.
- Removing extremely long sentence pairs.
- Removing rows with extreme source-target length ratio.
- Checking script consistency for Pashto and English text.

C. Dataset Cleaning Summary

VIII. Dataset Split

The cleaned dataset was divided into training, validation, and test splits. The training split contains 72,782 rows, while validation and test splits contain 9,098 rows each.

IX. Baseline Translation Model

A. Model Selection

The baseline model used in this project is:

`facebook/nllb-200-distilled-600M`

The model was selected because it supports multilingual translation across many languages, including Pashto, English, and Hindi. It provides a strong starting point for low-resource translation without training a model from scratch.

TABLE III
NLLB language codes used in this project.

Language	NLLB Code
Pashto	pbt_Arab
English	eng_Latn
Hindi	hin_Deva

TABLE IV
Standardized NLLB baseline evaluation.

Metric	Score
BLEU	19.63
chrF++	35.48
TER	75.68
COMET	0.6818

TABLE V
Average CPU inference time.

Translation Direction	Time per Sentence
Pashto-to-English	4.46 sec
Pashto-to-Hindi Direct	5.11 sec
Pashto-to-English-to-Hindi Pivot	4.76 sec

B. Language Codes

C. Translation Directions

The system supports:

- Pashto-to-English translation.
- Direct Pashto-to-Hindi translation.
- Pivot-based Pashto-to-English-to-Hindi translation.

X. Baseline Evaluation

A. Standardized Automatic Metrics

The earlier preliminary baseline values were replaced by a standardized evaluation computed from the same prediction and reference files used for all NLLB-based systems.

BLEU and chrF++ are higher-is-better metrics, whereas TER is lower-is-better. COMET provides a complementary neural evaluation signal. All reported systems use the same 100-sentence evaluation set.

B. Inference-Time Analysis

The previously measured CPU inference-time comparison is retained as an engineering observation. It is not treated as a translation-quality metric.

XI. Dataset Quality Integration

A. High-Quality Subset

A high-quality 10,000-pair subset was prepared from the cleaned dataset. This subset was selected for fine-tuning because training on a smaller but cleaner dataset can be more effective than training on a larger noisy dataset.

TABLE VI
Final dataset resources.

File	Rows	Purpose
cleaned_90k.csv	90,978	Clean corpus
train_high_quality_10k.csv	10,000	Fine-tuning data
gold_test_candidates_100.csv	100	Evaluation candidates

TABLE VII
Semantic filtering outputs.

Output	Purpose
Semantic similarity scores	Pair-level similarity scoring
Filtered 8000-pair dataset	Main filtered training subset
Top 2000 clean subset	Cleaner small training subset
Top 4000 clean subset	Cleaner medium training subset
Top examples preview	Manual inspection of best pairs

B. Gold Test Candidate Set

A 100-sample gold test candidate set was prepared for controlled evaluation. This test set allows direct comparison between the baseline and fine-tuned models.

C. Final Dataset Resources

XII. Semantic Similarity Filtering

Rule-based cleaning removes obvious noisy pairs, but it cannot always detect weak semantic alignment. Therefore, an additional semantic similarity filtering stage was added. The goal is to score Pashto-English sentence pairs using multilingual sentence embeddings and retain the most semantically aligned examples for future fine-tuning.

The semantic filtering process follows these steps:

- 1) Encode Pashto source sentences using a multilingual sentence embedding model.
- 2) Encode English target sentences using the same embedding model.
- 3) Compute cosine similarity between source and target embeddings.
- 4) Sort sentence pairs by semantic similarity score.
- 5) Retain the best aligned sentence pairs for fine-tuning.

$$\text{sim}(x, y) = \frac{x \cdot y}{\|x\| \|y\|} \quad (1)$$

A multilingual sentence-transformer model was used for this stage. The filtering stage generated pair-level semantic similarity scores, filtered training subsets, top high-similarity examples, and summary statistics.

A. Semantic Filtering Interpretation

The semantic filtering step improves the data-quality pipeline by ranking sentence pairs according to cross-lingual similarity. Higher similarity values indicate better alignment between the Pashto source sentence and the English reference sentence. Lower similarity values may indicate noisy, mismatched, incomplete, or weakly translated sentence pairs.

TABLE VIII
Original LoRA fine-tuning configuration.

Component	Configuration
Base model	NLLB 600M
Source language	pbt_Arab
Target language	eng_Latn
Fine-tuning method	LoRA
LoRA rank	8
LoRA alpha	16
LoRA dropout	0.05
Target modules	q_proj, v_proj
Training data	High-quality 10k subset
Evaluation data	100 gold candidates

This step is important in low-resource NMT because a smaller clean dataset can sometimes produce better fine-tuning results than a larger noisy dataset. The filtered 8000-pair dataset and the top 2000/top 4000 clean subsets were therefore prepared for improved LoRA experiments.

XIII. Qualitative Error Analysis Framework

Automatic metrics cannot fully explain omissions, hallucinations, named-entity errors, number errors, tense changes, literal translation, or poor fluency. The project therefore defines a qualitative analysis framework covering:

- missing or omitted information;
- incorrect or changed meaning;
- incomplete translation;
- named-entity and number errors;
- tense and grammatical mismatch;
- literal or unnatural translation;
- hallucinated content;
- untranslated text and repetition.

A structured evaluation template was prepared for future bilingual annotation. However, a formal human evaluation has not yet been completed. The examples reported in this paper are automatically selected comparison candidates and must not be interpreted as human preference judgments.

XIV. LoRA Fine-Tuning

A. Motivation

Full fine-tuning of NLLB is computationally expensive. LoRA is selected because it trains a small number of adapter parameters while freezing most of the base model. This reduces memory requirements and makes fine-tuning more practical.

B. Original LoRA Configuration

C. Semantic-Filtered LoRA Configuration

A semantic-filtered LoRA experiment was added after semantic similarity filtering. The semantic-filtered LoRA model was trained using the filtered Pashto-English subset and evaluated on the same gold test candidate set.

The long repository filenames were kept in the GitHub project for reproducibility, but shortened names are used in this paper to avoid layout overflow in IEEE two-column format.

TABLE IX
Semantic-filtered LoRA experiment.

Component	Configuration
Base model	NLLB 600M
Fine-tuning method	LoRA
Training data	Semantic-filtered 8000-pair subset
Evaluation data	100 gold test candidates
Output file	Semantic LoRA prediction CSV
Score file	Semantic LoRA BLEU/chrF score CSV

TABLE X
Standardized NLLB and LoRA results.

System	BLEU	chrF++	TER	COMET
Baseline NLLB	19.63	35.48	75.68	0.6818
Original LoRA	19.77	36.31	76.83	0.6775
Semantic LoRA	20.38	36.61	76.45	0.6819

XV. Fine-Tuning Results

All NLLB-based systems were re-evaluated using the same source sentences, English references, decoding outputs, and metric implementations.

Semantic-filtered LoRA achieved the highest BLEU and chrF++ point estimates among the NLLB variants. Relative to the pretrained baseline, it increased corpus BLEU by +0.75 and chrF++ by +1.12. However, COMET changed by only +0.000082, which is effectively a tie. Its TER was also higher than the baseline.

Original LoRA improved chrF++ but produced a slightly lower COMET score and a higher TER. The results therefore indicate modest and metric-dependent improvement rather than uniform superiority across all evaluation measures.

XVI. Improved LoRA Experiments

To improve translation quality beyond the initial LoRA results, additional experiments were designed using cleaner semantic-filtered data and stronger adapter settings.

A. Improved Data Selection

Two cleaner subsets were created from semantic similarity scores:

- `train_semantic_top2000_clean.csv`
- `train_semantic_top4000_clean.csv`

These subsets additionally apply sentence length and length-ratio filtering. The motivation is that smaller but cleaner data may outperform larger noisy data in low-resource settings.

B. Strong LoRA and Strong-v2 LoRA

A heavier LoRA setting was tested using more trainable modules. However, local CPU/RAM constraints made the heaviest setup difficult to run. Therefore, a safer strong-v2 configuration was prepared using attention projection modules only.

TABLE XI
Strong-v2 LoRA improvement setup.

Component	Configuration
LoRA rank	12
LoRA alpha	24
Target modules	q_proj, k_proj, v_proj, out_proj
Dropout	0.08
Training data	Semantic top-clean subset
Goal	Improve chrF while avoiding CPU/RAM failure

TABLE XII
Hindi comparison outputs.

Column	Purpose
Pashto source	Original input sentence
Pivot English	Intermediate English translation
Direct Hindi	Direct Pashto-to-Hindi output
Pivot Hindi	Pashto-to-English-to-Hindi output
Manual better method	Human comparison field
Manual comment	Error and quality notes

C. Improved Experiment Interpretation

The improved LoRA experiment tests whether stronger adapters and cleaner semantic-filtered data can improve translation quality beyond the original LoRA setup. A larger adapter can potentially learn better task-specific changes, but it also increases memory and compute requirements.

The strong-v2 configuration is therefore a practical compromise. It is stronger than the initial LoRA setup because it uses more attention projection modules, but it avoids the heavier feed-forward modules that caused local CPU/RAM failure. This makes it suitable for continued experimentation under limited hardware constraints.

XVII. Direct Hindi and Pivot Hindi Translation

The project explores Hindi translation through two approaches. The first approach is direct Pashto-to-Hindi translation using NLLB. The second approach is pivot translation, where Pashto is first translated into English and then English is translated into Hindi.

The pivot approach is promising because Pashto-to-English can be evaluated more reliably using English references. If the Pashto-to-English model improves, the downstream English-to-Hindi output can also improve.

A. Direct-vs-Pivot Hindi Comparison

A direct-vs-pivot Hindi comparison file was generated using the gold test candidates. Each sample contains the Pashto source, pivot English translation, direct Hindi output, pivot Hindi output, and manual review columns.

This comparison is necessary because Hindi output cannot be judged reliably using BLEU or chrF without gold Hindi references. Manual review is therefore required to decide whether direct or pivot translation is better.

TABLE XIII
Ablation study design.

ID	Model	Data	Purpose
E0	NLLB	None	Baseline
E1	NLLB+LoRA	100 pairs	Pipeline test
E2	NLLB+LoRA	500 pairs	Small training
E3	NLLB+LoRA	1k pairs	Scaling test
E4	NLLB+LoRA	3k pairs	Medium training
E5	NLLB+LoRA	10k pairs	Main run
E6	NLLB+LoRA	Semantic 8k	Filtering test
E7	Strong-v2 LoRA	Top semantic 2k	Adapter test
E8	Strong-v2 LoRA	Top semantic 4k	Larger clean-data test

XVIII. IndicTrans2 Applicability and Status

IndicTrans2 was considered for the English-to-Hindi stage of the pivot translation pipeline:

Pashto → English using NLLB/LoRA → Hindi using IndicTrans2.

IndicTrans2 is not used as a direct Pashto-to-English baseline in this study. Native Windows installation of the official processing toolkit was not completed because its Cython extension required an unavailable compiler toolchain. The quantitative Pashto-to-English architecture comparison was therefore conducted using NLLB, M2M100, and mBART-50.

Until IndicTrans2 outputs are generated in a supported Linux or Colab environment, it is reported only as an applicable future English-to-Hindi pivot experiment. No IndicTrans2 quality result is claimed in this paper.

XIX. Ablation Study

A complete ablation study is needed to understand how data size, semantic filtering, and LoRA settings affect translation performance.

XX. Reproducibility

The repository contains code for dataset processing, semantic filtering, fine-tuning, evaluation, checkpoint comparison, Hindi comparison, and report generation. The important files include:

- `src/finetune_nllb_lora.py`
- `src/finetune_nllb_lora_strong_v2.py`
- `src/week5_semantic_filter.py`
- `src/evaluate_week4_model.py`
- `src/evaluate_improved_generation.py`
- `src/compare_remaining_checkpoints.py`
- `src/remaining_direct_vs_pivot_hindi.py`
- `data/train_high_quality_10k.csv`
- `data/train_semantic_filtered_8000.csv`
- `data/gold_test_candidates_100.csv`

Large model checkpoints are not stored in the repository because they are too large for normal GitHub tracking. Instead, scripts, evaluation outputs, graphs, and reports are pushed to maintain reproducibility.

XXI. Limitations

The present study has the following limitations:

- The standardized evaluation currently contains 100 shared Pashto–English examples.
- Full three-seed training is not yet complete; seed configurations have been prepared but must still be executed.
- M2M100 and mBART-50 are evaluated zero-shot and have not yet received equal-budget LoRA adaptation.
- The qualitative examples are automatically selected and have not been validated by multiple bilingual annotators.
- BLEU, chrF++, TER, and COMET provide complementary but imperfect quality estimates.
- Hindi evaluation lacks a gold Hindi reference set.
- IndicTrans2 inference was not completed in the native Windows environment.
- CPU-only training limits the number of model, data-size, seed, and hyperparameter experiments.
- The current results do not establish state-of-the-art performance.

XXII. Extended Evaluation and Instructor-Requested Improvements

A. Informative Research Pipeline

The original screenshot-style summary figures were replaced by an informative end-to-end research diagram connecting corpus preparation, leakage control, semantic filtering, LoRA adaptation, external baselines, automatic metrics, qualitative analysis, and Hindi translation.

B. Multilingual Architecture Comparison

To place NLLB in a broader multilingual context, M2M100 and mBART-50 were evaluated zero-shot on the same Pashto–English instances [9], [10].

C. Qualitative Translation Examples

The instructor-requested qualitative examples compare the English reference, pretrained NLLB output, original LoRA output, and semantic-filtered LoRA output. Improvement and regression candidates were selected using sentence-level chrF++ differences. They illustrate model behaviour but do not replace bilingual human evaluation.

D. Statistical Analysis

For Semantic LoRA relative to the NLLB baseline, the implemented paired sentence-level bootstrap analysis reported:

- BLEU: difference +1.8850, $p = 0.0328$;
- chrF++: difference +1.1512, $p = 0.1148$;
- TER: difference +2.7728, $p = 0.6112$;
- COMET: difference +0.000082, $p = 0.9744$.

Only the sentence-level BLEU comparison crossed the 0.05 threshold in the pilot analysis. chrF++, TER, and COMET did not provide statistically significant evidence of general superiority. Corpus-level metric differences and averages of sentence-level metric differences are reported as distinct quantities.

Extended Quality-Aware Pashto NMT Research Pipeline

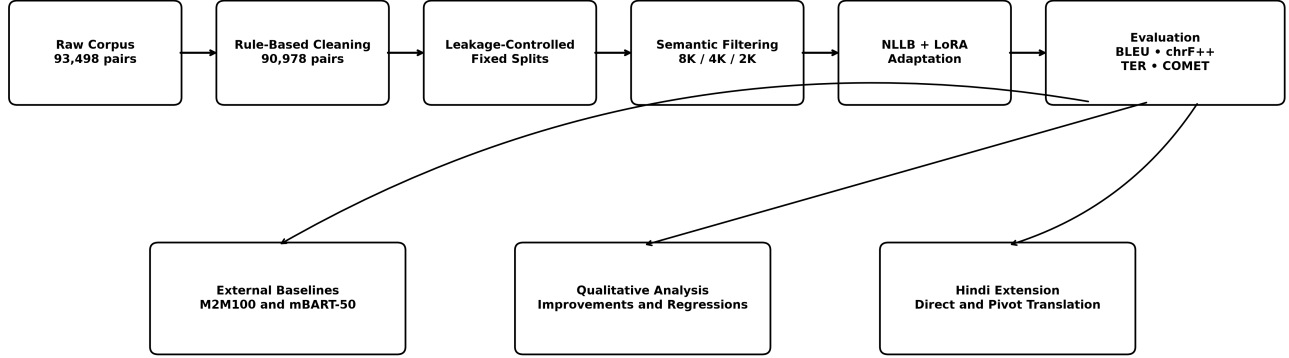


Fig. 1. Extended quality-aware Pashto NMT research pipeline.

TABLE XIV
Standardized Pashto–English evaluation across NLLB, LoRA, and external multilingual baselines.

Model	BLEU	chrF++	TER	COMET	Samples	Setting
Baseline NLLB	19.63	35.48	75.68	0.6818	100	NLLB baseline or LoRA adaptation
Original LoRA	19.77	36.31	76.83	0.6775	100	NLLB baseline or LoRA adaptation
Semantic LoRA	20.38	36.61	76.45	0.6819	100	NLLB baseline or LoRA adaptation
M2M100 418M	14.62	30.59	85.42	0.6169	100	External zero-shot baseline
mBART-50	22.65	38.08	75.77	0.6932	100	External zero-shot baseline

TER is lower-is-better. M2M100 and mBART-50 are external zero-shot baselines and were not fine-tuned on the project corpus.

TABLE XV
Representative qualitative comparison of the pretrained NLLB baseline, original LoRA, and semantic-filtered LoRA outputs.

Example 1: Improvement candidate

Pashto source	په مادي توگه د فرعي لیستونو جوړولو لپاره برخه.
Reference	Segment manually to create sub-lists.
NLLB baseline	Physically part to form sub-lists.
Original LoRA	Physically part to create sub-lists.
Semantic LoRA	Physically part to create sub-lists.
chrF++ change	+24.31 (Semantic LoRA minus baseline)

Example 2: Improvement candidate

Pashto source	د بحث سره یوځای غواړئ؟
Reference	Want to join the discussion?
NLLB baseline	Would you like to join the discussion?
Original LoRA	Would you like to join the discussion?
Semantic LoRA	Want to join the discussion?
chrF++ change	+23.76 (Semantic LoRA minus baseline)

Example 3: Regression candidate

Pashto source	1. د ورکړې شرایط؟
Reference	1. Payment Terms ?
NLLB baseline	1. Payment terms?
Original LoRA	1. Conditions of payment?
Semantic LoRA	1. Conditions of payment?
chrF++ change	-36.26 (Semantic LoRA minus baseline)

Examples were selected by sentence-level chrF++ change and illustrate both improvement and regression behavior; they are not substitutes for bilingual human evaluation.

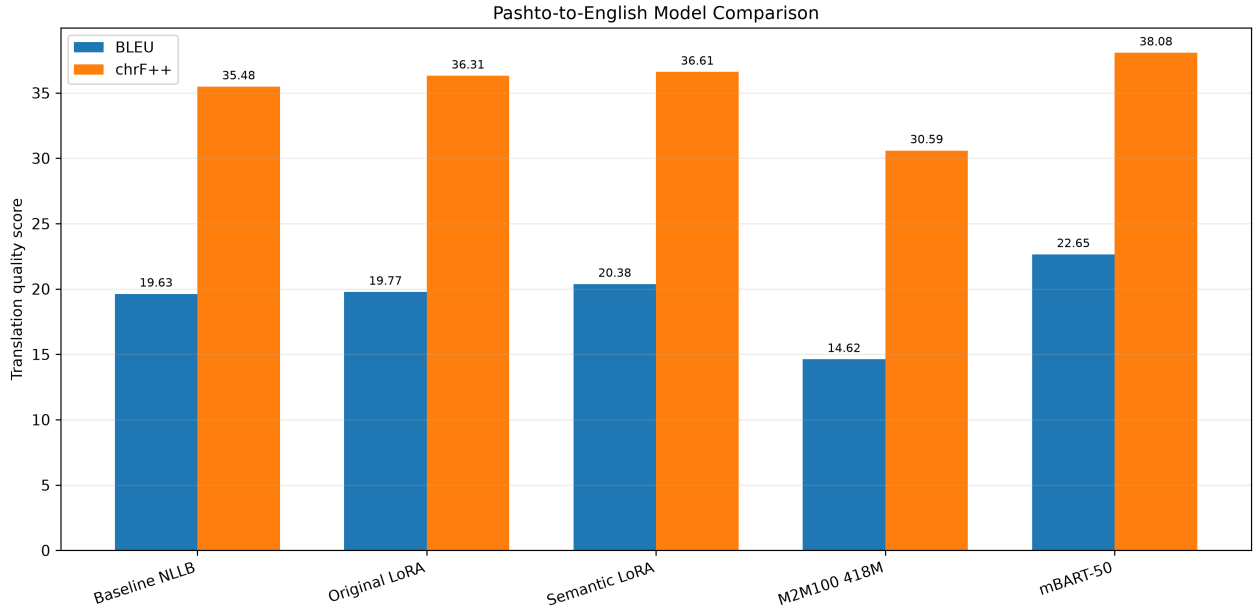


Fig. 2. BLEU and chrF++ comparison across the evaluated NLLB, LoRA, M2M100, and mBART-50 systems.

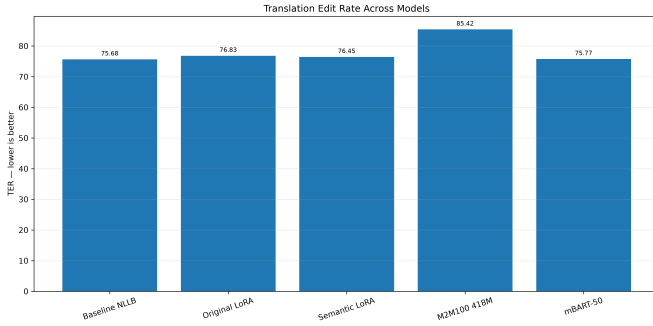


Fig. 3. Translation Edit Rate comparison. Lower values are better.

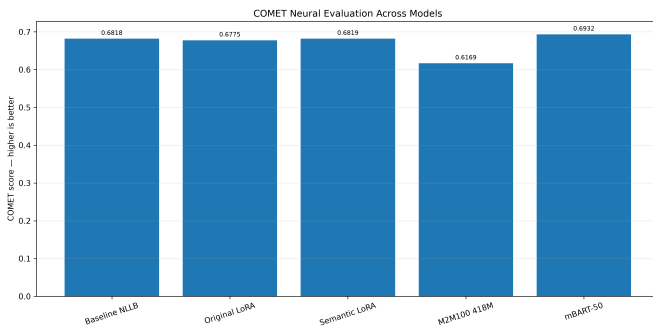


Fig. 4. COMET neural evaluation for systems with completed COMET scores.

E. Additional Completed Engineering Work

Beyond the instructor’s requested figures, qualitative examples, and model comparisons, the extended project also completed:

- standardized BLEU, chrF++, TER, and COMET evalua-

tion;

- pairwise wins, losses, and ties;
- automatic translation diagnostics;
- fixed source-grouped train, validation, and test splits;
- zero exact-pair and Pashto-source overlap;
- SHA-256 dataset checksums;
- package and runtime environment records;
- machine-readable training manifests;
- configurations for seeds 13, 42, and 2026;
- a successful CPU LoRA smoke-training run;
- successful adapter saving, reloading, and translation-generation checks.

The smoke experiment is an engineering validation and is not included as a final translation-quality result.

XXIII. Discussion

The expanded evaluation changes the interpretation of the initial experiments. Semantic-filtered LoRA obtains the best BLEU and chrF++ point estimates among the NLLB variants, but its almost unchanged COMET score indicates that the improvement is not uniformly reflected by a neural semantic metric. Its higher TER also shows that better character-level overlap does not necessarily imply fewer reference edits.

The M2M100 and mBART-50 comparisons reduce dependence on a single multilingual architecture. However, because these systems are evaluated zero-shot, the comparison measures pretrained transfer quality rather than equal-budget adaptation. A stronger experiment should apply the same data size, LoRA configuration, training steps, decoding settings, and seeds to every compatible model.

The qualitative examples make the metric results easier to interpret by exposing both improvement and regression candidates. They must be reviewed by bilingual annotators

before being described as improvements in human-perceived adequacy or fluency.

The present contribution is therefore best described as a reproducible empirical analysis of semantic filtering for parameter-efficient Pashto–English adaptation. The current evidence does not justify claims of state-of-the-art performance or universal statistically significant superiority.

XXIV. Conclusion

This paper presented a quality-aware and reproducible pipeline for low-resource Pashto neural machine translation. Starting from 93,498 raw Pashto–English sentence pairs, rule-based cleaning retained 90,978 pairs, after which semantic filtering and LoRA adaptation were applied.

On the standardized 100-sentence evaluation set, pretrained NLLB achieved BLEU 19.63, chrF++ 35.48, TER 75.68, and COMET 0.6818. Original LoRA achieved BLEU 19.77 and chrF++ 36.31, while semantic-filtered LoRA achieved the strongest NLLB-variant BLEU and chrF++ scores of 20.38 and 36.61. Its COMET score of 0.6819 remained effectively tied with the pretrained baseline.

The revised study additionally contributes an informative pipeline diagram, model-level quality figures, qualitative translation examples, M2M100 and mBART-50 comparisons, paired bootstrap analysis, automatic diagnostics, leakage-controlled splits, dataset hashes, training manifests, and multi-seed experiment configurations. IndicTrans2 is retained only for the applicable future English-to-Hindi pivot stage.

Future work should complete the full multi-seed runs, expand evaluation to a substantially larger independent test set, apply equal-budget adaptation to a second model family, and conduct blinded bilingual human evaluation. These steps are required before making strong claims about robustness, significance, and generalization.

Repository

The complete project repository is available at:

<https://github.com/Hrithikcrick/pashto-nmt-carnot-internship>

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